

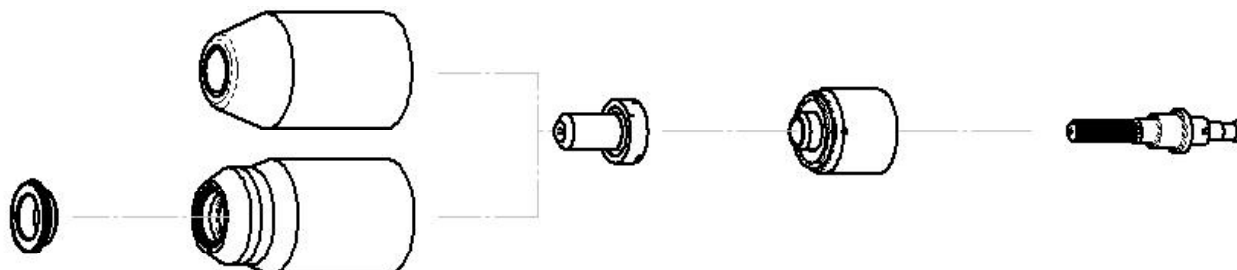
CUTMASTER A40

4T.08 Recommended Cutting Speeds for Machine and Automated Torches With Exposed Tip

Mild Steel

40A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.036	70 (25') 75 (50')	101	0.14	160	0.18	0.0	0.05
16		0.060		103	0.14	140	0.18	0.0	0.05
14		0.075		105	0.14	120	0.18	0.1	0.06
12		0.105		108	0.14	80	0.18	0.2	0.06
10		0.135		110	0.14	60	0.18	0.3	0.06
	3/16	0.188		111	0.14	55	0.18	0.4	0.06
	1/4	0.250		117	0.14	40	0.18	0.5	0.07
	3/8	0.375		119	0.14	25	0.18	1.2	0.07
	1/2	0.500		120	0.14	15	0.2	2.0	0.07

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	4.8 (7.6m) 5.2 (15.2m)	101	3.6	3990	4.6	0.0	1.1
2		105	3.6	2920	4.6	0.1	1.4
3		109	3.6	1810	4.6	0.3	1.5
4		110	3.6	1470	4.6	0.3	1.6
5		112	3.6	1345	4.6	0.4	1.6
6		116	3.6	1100	4.6	0.5	1.7
8		118	3.6	815	4.6	1.0	1.7
10		119	3.6	595	4.6	1.5	1.8
12		120	3.6	435	5.1	2.0	1.8

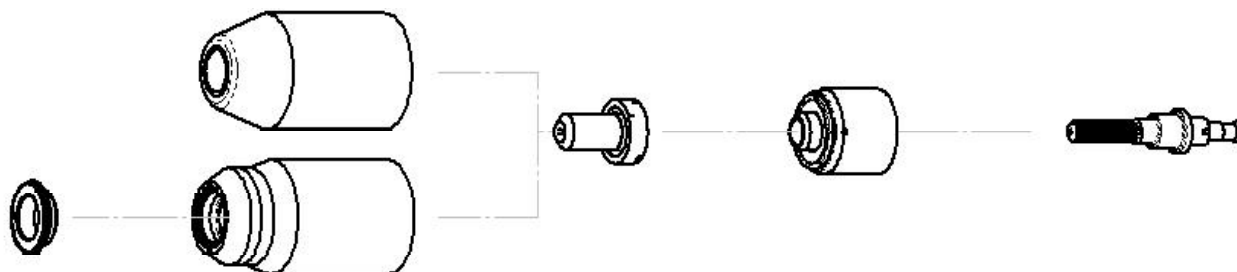
BOLD TYPE indicates maximum piercing parameters.

CUTMASTER A40

Stainless Steel

40A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
18		0.050	75 (25') 80 (50')	110	0.19	60	0.20	0.00	0.07
16		0.063		100	0.19	50	0.20	0.00	0.07
14		0.078		105	0.19	45	0.20	0.10	0.07
12		0.109		110	0.19	40	0.20	0.20	0.07
10		0.141		108	0.19	35	0.20	0.30	0.07
	3/16	0.188		110	0.19	30	0.20	0.40	0.07
	1/4	0.250		120	0.19	18	0.20	0.50	0.08
	3/8	0.375		126	0.19	10	0.20	2.00	0.08
	1/2	0.500		118	0.19	8	Edge start		0.09

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.2 (7.6) 5.5 (15.2)	112	4.8	1670	4.8	0.0	1.7
2		105	4.8	1140	5.1	0.1	1.8
3		109	4.8	980	5.1	0.2	1.8
4		109	4.8	845	5.1	0.3	1.8
5		111	4.8	715	5.1	0.4	1.8
6		118	4.8	525	5.1	0.5	2.0
8		123	4.8	350	5.1	1.5	2.0
10		125	4.8	245	5.1	2.0	2.0
12		120	4.8	215	<i>Edge Start</i>		2.2

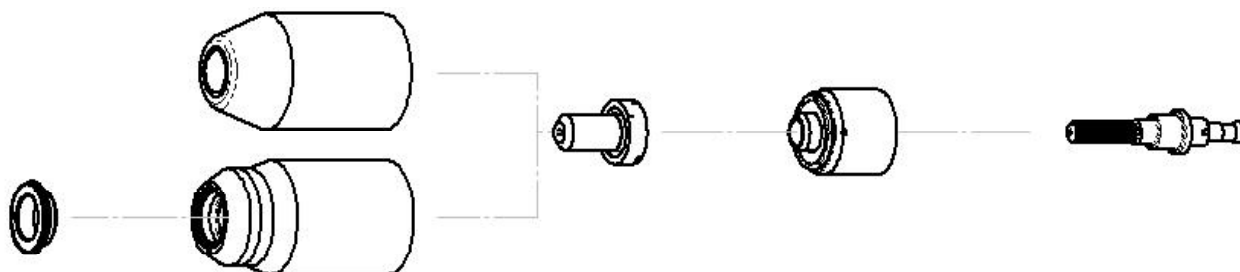
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A40

Aluminum

40A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.040	70 (25') 75 (50')	95	0.16	300	0.16	0.0	0.06
16		0.063		97	0.16	170	0.16	0.0	0.06
12		0.097		113	0.16	100	0.16	0.2	0.07
11		0.125		115	0.16	90	0.18	0.3	0.07
9		0.160		113	0.18	85	0.18	0.4	0.07
	3/16	0.188		116	0.18	75	0.18	0.5	0.07
	1/4	0.250		128	0.18	30	0.18	1.0	0.08
	3/8	0.375		150	0.18	10	Edge Start		0.09

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1.0	4.8 (7.6m) 5.2 (15.2m)	95	4.1	7620	4.1	0.0	1.5
2.0		104	4.1	3500	4.1	0.2	1.6
3.0		115	4.1	2350	4.6	0.3	1.7
4.0		113	4.6	2170	4.6	0.4	1.7
5.0		118	4.6	1740	4.6	0.5	1.8
6.0		125	4.6	1015	4.6	0.8	1.9
8.0		139	4.6	500	<i>Edge Start</i>		2.0
10.0		153	4.6	180	<i>Edge Start</i>		2.2

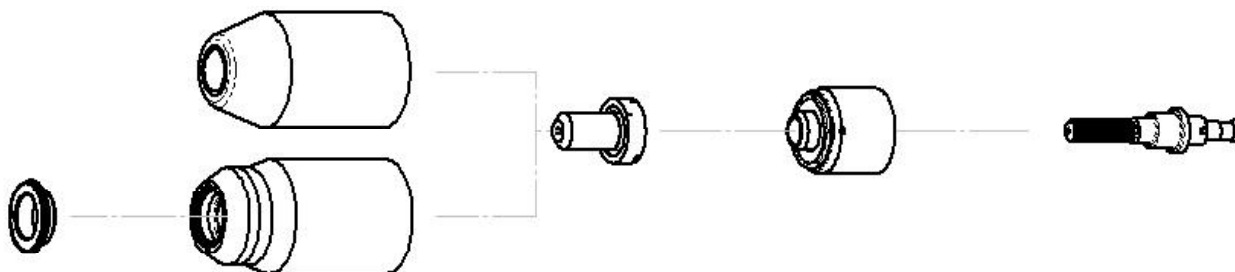
BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

CUTMASTER A40

Mild Steel

60A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8210	9-8213	9-8232

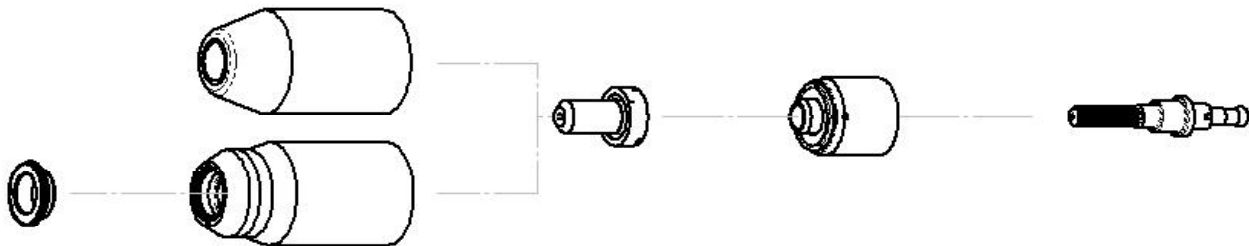
Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.060	85 (25') 90 (50')	118	0.19	290	0.19	0.00	0.08
14		0.075		120	0.19	285	0.19	0.10	0.08
11		0.120		118	0.19	180	0.19	0.10	0.08
10		0.135		119	0.19	176	0.19	0.10	0.08
	3/16	0.188		121	0.19	100	0.19	0.20	0.08
	1/4	0.250		122	0.19	80	0.19	0.30	0.08
	3/8	0.375		124	0.19	50	0.19	0.50	0.09
	1/2	0.500		132	0.19	26	0.19	0.75	0.10
	5/8	0.625		135	0.19	19	Edge Start		0.10
	3/4	0.750		136	0.19	14	Edge Start		0.08
	1	1.000		150	0.19	6	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	115	4.8	7540	4.8	0	2.1
2		120	4.8	7015	4.0	0.10	1.9
3		118	4.8	4570	4.8	0.10	0.1
4		120	4.8	3650	4.8	0.20	1.9
5		121	4.8	2465	4.8	0.20	2.1
6		122	4.8	2145	4.8	0.30	2.0
8		123	4.8	1635	4.8	0.40	2.2
10		125	4.8	1180	4.8	0.60	2.4
12		130	4.8	795	4.8	0.75	2.4
15		134	4.8	530	<i>Edge Start</i>		2.4
20		138	4.8	325	<i>Edge Start</i>		2.2
25		149	4.8	165	<i>Edge Start</i>		2.7

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A40

Stainless Steel 60A Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.063	85 (25') 90 (50')	119	0.19	350	0.20	0.00	0.05
14		0.078		116	0.19	300	0.20	0.10	0.07
11		0.125		123	0.19	150	0.20	0.10	0.07
10		0.141		118	0.19	125	0.20	0.10	0.08
	3/16	0.188		122	0.19	90	0.20	0.20	0.08
	1/4	0.250		123	0.19	65	0.20	0.30	0.09
	3/8	0.375		130	0.19	30	0.20	0.50	0.09
	1/2	0.500		132	0.19	21	0.20	0.90	0.08
	5/8	0.625		132	0.19	14	Edge Start		0.11
	3/4	0.750		135	0.19	10	Edge Start		0.10

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	124	4.8	10890	5.1	0.00	0.8
2		116	4.8	7560	5.1	0.10	1.7
3		122	4.8	4365	5.1	0.10	1.7
4		119	4.8	2865	5.1	0.20	2.1
5		122	4.8	2195	5.1	0.20	2.1
6		123	4.8	1790	5.1	0.30	2.2
8		127	4.8	1190	5.1	0.40	2.2
10		130	4.8	725	5.1	0.50	2.2
12		132	4.8	580	5.1	0.90	2.1
15		132	4.8	405	Edge Start		2.6
20		136	4.8	230	Edge Start		2.5

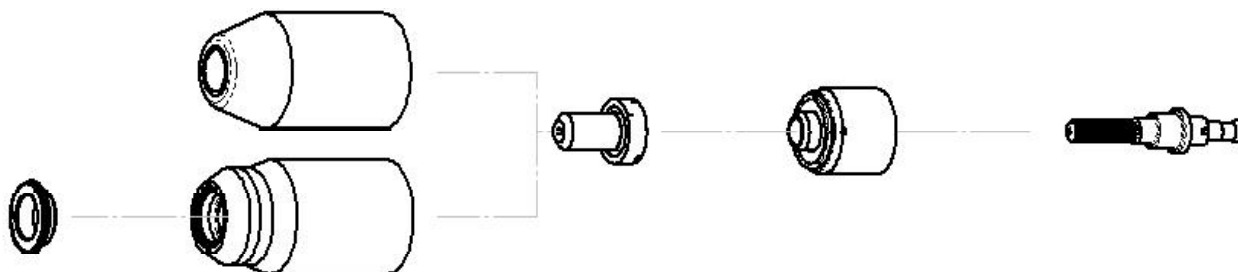
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A40

Aluminum

60A

Air Plasma / Air Shield



Deflector	Standard Shield Cup Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8243	9-8218 9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.064	85 (25') 90 (50')	123	0.25	440	0.25	0.00	0.08
14		0.079		126	0.25	300	0.25	0.10	0.09
11		0.120		128	0.25	250	0.25	0.10	0.09
	3/16	0.188		132	0.25	170	0.25	0.20	0.09
	1/4	0.250		132	0.25	85	0.25	0.30	0.09
	3/8	0.375		141	0.25	45	0.25	0.50	0.10
	1/2	0.500		148	0.25	30	0.25	0.80	0.09
	5/8	0.625		145	0.19	18	Edge Start		0.08
	3/4	0.750		147	0.19	12	Edge Start		0.10

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	118	6.4	17010	6.4	0.00	1.8
2		126	6.4	7680	6.4	0.10	2.3
3		128	6.4	6410	6.4	0.10	2.3
4		130	6.4	5230	6.4	0.20	2.3
5		132	6.4	4010	6.4	0.20	2.4
6		132	6.4	2640	6.4	0.30	2.4
8		137	6.4	1630	6.4	0.40	2.4
10		142	6.4	1085	6.4	0.60	2.4
12		146	6.4	845	6.4	0.70	2.3
15		146	4.8	540	<i>Edge Start</i>		2.1
20		148	4.8	260	<i>Edge Start</i>		2.5

BOLD TYPE indicates maximum piercing parameters. ***BOLD ITALIC*** indicates edge starts only.

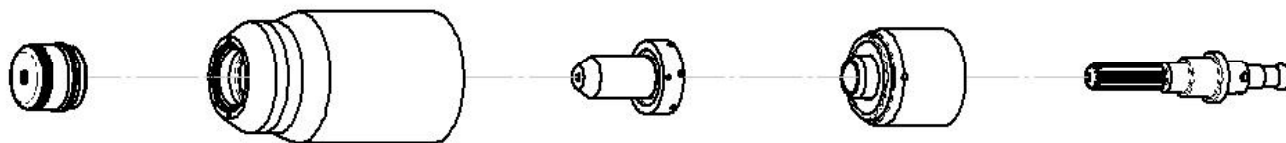
CUTMASTER A40

4T.09 Recommended Cutting Speeds for Machine and Automated Torches With Shielded Tip

Mild Steel

40A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8245	9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.036	75 (25") 80 (50")	107	0.16	130	0.16	0.0	0.06
16		0.060		113	0.16	120	0.16	0.0	0.06
14		0.075		108	0.16	90	0.16	0.1	0.06
12		0.105		111	0.16	75	0.16	0.2	0.06
10		0.135		114	0.16	65	0.16	0.3	0.07
	3/16	0.188		115	0.16	65	0.16	0.4	0.07
	1/4	0.250		118	0.16	45	0.16	0.5	0.07
	3/8	0.375		123	0.16	23	0.16	1.2	0.08
	1/2	0.500		128	0.16	18	0.16	2.0	0.08

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.2 (7.6) 5.5 (15.2)	108	4.1	3266	4.1	0.0	1.4
2		108	4.1	2239	4.1	0.0	1.5
3		112	4.1	1794	4.1	0.1	1.7
4		114	4.1	1651	4.1	0.2	1.7
5		115	4.1	1578	4.1	0.3	1.7
6		117	4.1	1256	4.1	0.4	1.7
8		121	4.1	853	4.1	0.5	1.7
10		124	4.1	565	4.1	1.2	1.8
12		127	4.1	485	4.1	2.0	1.9

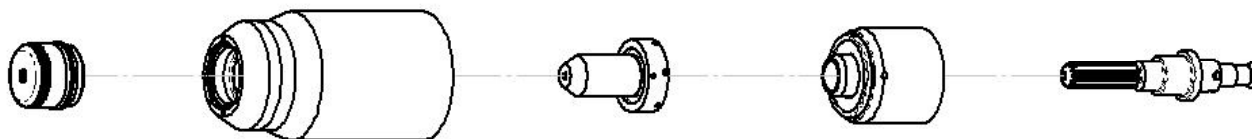
BOLD TYPE indicates maximum piercing parameters.

CUTMASTER A40

Stainless Steel

40A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8245	9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
18		0.050	75 (25') 80 (50')	110	0.16	60	4.1	0.0	0.06
16		0.063		108	0.16	50	4.1	0.1	0.07
14		0.078		114	0.16	45	4.1	0.1	0.07
12		0.109		113	0.16	40	4.1	0.2	0.07
10		0.141		116	0.16	35	4.1	0.3	0.07
	3/16	0.188		115	0.16	30	4.1	0.4	0.07
	1/4	0.250		118	0.16	20	4.1	1.2	0.07
	3/8	0.375		125	0.16	15	4.1	1.8	0.08
	1/2	0.500		127	0.16	10	4.1	2.0	0.08

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.2 (7.6) 5.5 (15.2)	109	4.1	1670	4.1	0.0	1.7
2		114	4.1	1140	4.1	0.1	1.8
3		114	4.1	980	4.1	0.2	1.8
4		116	4.1	845	4.1	0.3	1.8
5		115	4.1	725	4.1	0.4	1.8
6		117	4.1	565	4.1	0.5	2.0
8		122	4.1	440	4.1	1.5	2.0
10		125	4.1	360	4.1	1.8	2.0
12		127	4.1	280	4.1	2.0	2.2

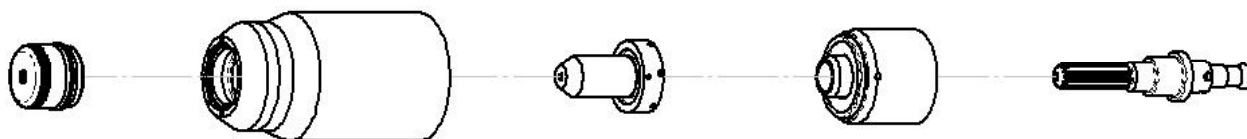
BOLD TYPE indicates maximum piercing parameters.

CUTMASTER A40

Aluminum

40A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8245	9-8237	9-8208	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
20		0.040	75 (25') 80 (50')	110	0.19	300	0.20	0.0	0.06
16		0.063		113	0.19	170	0.20	0.1	0.07
12		0.097		120	0.19	100	0.20	0.2	0.07
11		0.125		125	0.19	90	0.20	0.3	0.07
9		0.160		126	0.19	85	0.20	0.4	0.07
	3/16	0.188		128	0.19	70	0.20	0.5	0.07
	1/4	0.250		137	0.19	30	0.20	1.0	0.08
	3/8	0.375		145	0.19	10	0.20	2.0	0.09

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1.0	5.2 (7.6) 5.5 (15.2)	110	4.8	7660	5.1	0.0	1.6
2.0		116	4.8	3490	5.1	0.2	1.8
3.0		124	4.8	2350	5.1	0.3	1.8
4.0		126	4.8	2170	5.1	0.4	1.8
5.0		129	4.8	1630	5.1	0.6	1.8
6.0		135	4.8	990	5.1	1.0	1.9
8.0		141	4.8	500	5.1	1.6	2.0
10.0		146	4.8	180	5.1	2.1	2.3

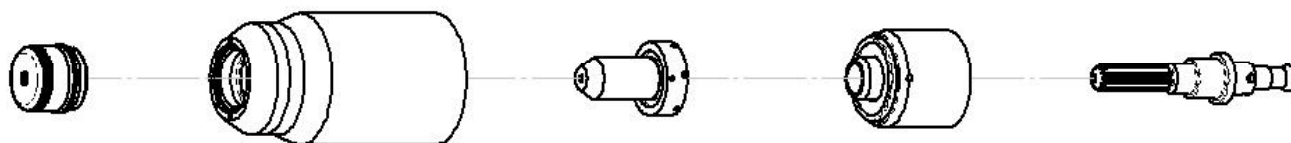
BOLD TYPE indicates maximum piercing parameters.

CUTMASTER A40

Mild Steel

60A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8238	9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.060	85 (25') 90 (50')	128	0.19	250	0.20	0.00	0.09
14		0.075		129	0.19	237	0.20	0.10	0.09
11		0.120		126	0.19	200	0.20	0.10	0.08
10		0.135		128	0.19	142	0.20	0.10	0.09
	3/16	0.188		132	0.19	115	0.20	0.20	0.09
	1/4	0.250		134	0.19	80	0.20	0.30	0.08
	3/8	0.375		132	0.19	34	0.20	0.70	0.10
	1/2	0.500		144	0.19	23	0.20	0.75	0.10
	5/8	0.625		150	0.19	14	Edge Start		0.10
	3/4	0.750		145	0.19	14	Edge Start		0.10
	1	1.000		160	0.19	4	Edge Start		0.12

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	127	4.8	6804	5.1	0	2.2
2		129	4.8	5942	5.1	0.10	2.2
3		126	4.8	5080	5.1	0.10	0.1
4		130	4.8	3316	5.1	0.20	2.3
5		132	4.8	2794	5.1	0.20	2.2
6		134	4.8	2230	5.1	0.30	2.1
8		133	4.8	1425	5.1	0.40	2.3
10		134	4.8	822	5.1	0.60	2.5
12		141	4.8	646	5.1	0.75	2.5
15		148	4.8	419	Edge Start		2.6
20		147	4.8	318	Edge Start		2.7
25		159	4.8	118	Edge Start		3.0

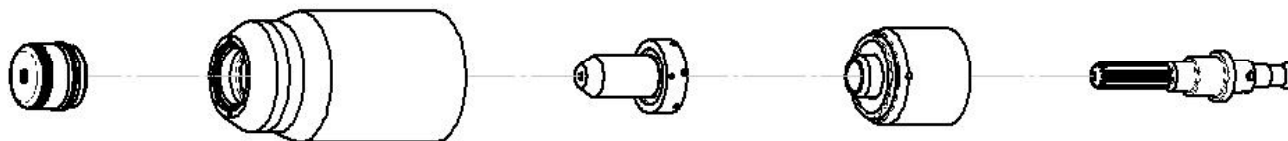
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A40

Stainless Steel

60A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8238	9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.063	85 (25') 90 (50')	110	0.13	165	0.20	0.00	0.06
14		0.078		116	0.13	155	0.20	0.10	0.07
11		0.125		118	0.13	125	0.20	0.10	0.07
10		0.141		126	0.13	80	0.20	0.10	0.09
	3/16	0.188		125	0.13	75	0.20	0.20	0.09
	1/4	0.250		127	0.13	60	0.20	0.30	0.08
	3/8	0.375		134	0.13	28	0.20	0.50	0.08
	1/2	0.500		136	0.19	17	0.25	0.75	0.09
	5/8	0.625		131	0.13	14	Edge Start		0.08
	3/4	0.750		142	0.19	10	Edge Start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	101	3.3	4590	5.1	0.00	1.2
2		116	3.3	3925	5.1	0.10	1.7
3		118	3.3	3285	5.1	0.10	1.7
4		126	3.3	1985	5.1	0.20	2.2
5		125	3.3	1850	5.1	0.20	2.1
6		127	3.3	1605	5.1	0.30	2.1
8		131	3.3	1100	5.1	0.40	2.1
10		134	3.3	670	5.1	0.50	2.1
12		136	4.8	490	6.4	0.90	2.2
15		132	3.3	375	Edge Start		2.6
20		144	4.8	230	Edge Start		2.5

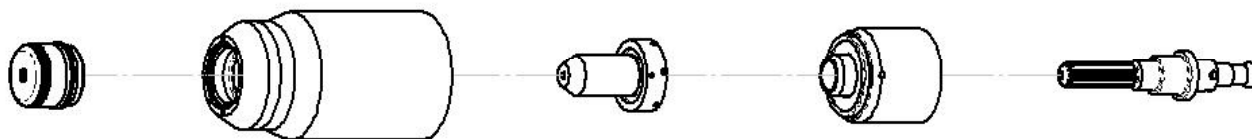
BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

CUTMASTER A40

Aluminum

60A

Air Plasma / Air Shield



Shield Cap	Maximum Life Shield Cup	Tip	Starter Cartridge	Electrode
9-8238	9-8237	9-8210	9-8213	9-8232

Material Thickness			Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(GA)	(in)	inch	PSI (torch lead length)	Volts	(in)	(ipm)	(in)	(sec)	(in)
16		0.064	85 (25') 90 (50')	118	0.13	350	0.20	0.00	0.07
14		0.079		118	0.13	350	0.20	0.10	0.08
11		0.120		123	0.13	275	0.20	0.10	0.08
	3/16	0.188		125	0.13	140	0.20	0.20	0.08
	1/4	0.250		132	0.13	80	0.20	0.30	0.08
	3/8	0.375		135	0.13	45	0.20	0.50	0.08
	1/2	0.500		140	0.13	26	0.20	0.80	0.08
	5/8	0.625		148	0.19	18	Edge start		0.10
	3/4	0.750		155	0.19	12	Edge start		0.11

Material Thickness	Gas Pressure (Air)	Arc Voltage	Torch Working Height	Travel Speed	Initial Piercing Height	Pierce Delay	Kerf Width @ Rec. Speed
(mm)	Bar (torch lead length)	Volts	(mm)	(mm/min)	(mm)	(sec)	(mm)
1	5.9 (7.6m) 6.2 (15.2m)	118	6.4	8890	6.4	0.00	2
2		118	6.4	8890	6.4	0.10	2.0
3		123	6.4	7070	6.4	0.10	2.1
4		124	6.4	5095	6.4	0.20	2.0
5		126	6.4	3335	6.4	0.20	2.0
6		130	6.4	2370	6.4	0.30	2.0
8		134	6.4	1570	6.4	0.40	2.0
10		136	6.4	1070	6.4	0.60	2.0
12		139	6.4	765	6.4	0.70	2.0
15		146	4.8	515	Edge start		2.4
20		157	4.8	260	Edge start		2.7

BOLD TYPE indicates maximum piercing parameters. **BOLD ITALIC** indicates edge starts only.

PATENT INFORMATION**Plasma Cutting Torch Patents**

The following parts are covered under U.S. and Foreign Patents as follows:

Catalog #	Description	Patent(s)
9-8215	Electrode	US Pat No(s) 6163008; 6987238 Other Pat(s) Pending
9-8232	Electrode	US Pat No(s) 6163008; 6987238 Other Pat(s) Pending
9-8213	Cartridge	US Pat No(s) 6903301; 6717096; 6936786; 6703581; D496842; D511280; D492709; D499620; D504142 Other Pat(s) Pending
9-8205	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8206	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8207	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8252	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8208	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8209	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8210	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8231	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8211	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8212	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8253	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8233	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8225	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8226	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8227	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8228	Tip	US Pat No(s) 6774336; 7145099; 6933461 Other Pat(s) Pending
9-8241	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8243	Shield Cap	US Pat No(s) 6914211; D493183 Other Pat(s) Pending
9-8235	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8236	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8237	Shield Cup	US Pat No(s) 6914211; D501632; D511633 Other Pat(s) Pending
9-8238	Shield Cap	US Pat No(s) 6914211; D496951 Other Pat(s) Pending

Catalog #	Description	Patent(s)
9-8239	Shield Cap	US Pat No(s) 6914211; D496951 Other Pat(s) Pending
9-8244	Shield Cap	US Pat No(s) 6914211; D505309 Other Pat(s) Pending
9-8245	Shield Cap	US Pat No(s) 6914211; D496951 Other Pat(s) Pending

The following parts are also licensed under U.S. Patent No. 5,120,930 and 5,132,512:

Catalog #	Description
9-8235	Shield Cap
9-8236	Shield Cap
9-8237	Shield Cup
9-8238	Shield Cap
9-8239	Shield Cap
9-8244	Shield Cap
9-8245	Shield Cap